

Which ANDES Converter Model for Virtual Inertia Scheduling?

REGCV1 vs. the REGF Grid-Forming Family — a Model-Selection Note

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Abstract

Virtual inertia scheduling (VIS) co-optimises generation dispatch with the *virtual inertia* M and *damping* D provided by grid-forming inverters. The choice of converter dynamic model therefore determines whether the scheduler’s decision variables even exist in the simulator. This note compares the seven renewable/converter models in ANDES 1.9.3 and concludes—from the model equations, the physical (swing-equation) requirement, and comparability with the foundational VIS literature—that **REGCV1** is the correct model for this study. **REGCV2** is an equivalent alternative; the EPRI grid-forming **REGF2** (VSM) is the natural realism upgrade for a robustness study; **REGCA1** (grid-following) and **REGF1/REGF3** (droop/dVOC) are unsuitable because they do not expose a schedulable inertia constant.

1 The requirement: what VIS actually schedules

VIS treats the emulated inertia M and damping D of each grid-forming IBR as *decision variables*, alongside generation dispatch, under frequency-security limits. The physical target being emulated is the synchronous-machine swing equation (per unit, ω in p.u.):

$$2H \frac{d\omega}{dt} = P_m - P_e - D(\omega - 1), \quad (1)$$

whose immediate consequence, for an imbalance ΔP , is the rate-of-change-of-frequency

$$\dot{f}_0 = \frac{f_0 \Delta P}{2H} = \frac{f_0 \Delta P}{M}, \quad M \equiv 2H. \quad (2)$$

A suitable model must therefore (i) be *grid-forming* (a controlled voltage source that sets frequency, not a current source that follows it), and (ii) expose an explicit *inertia constant* $M=2H$ (and damping D) as a tunable parameter. Droop gain alone is not inertia: it shapes the steady-state power split, not the $d\omega/dt$ transient that (2) governs.

2 The ANDES model landscape

ANDES 1.9.3 provides seven models in the **RenGen** group. Table 1 classifies them by control paradigm and, crucially, by whether an inertia constant is an explicit parameter.

3 What REGCV1 actually solves

REGCV1 is documented as a “voltage-controlled VSC with VSG control: double-loop PI control and swing-equation-based VSG control.” Its dynamic core (from the ANDES source) is, for the

Table 1: Renewable/converter models in ANDES 1.9.3 (RenGen group).

Model	Control paradigm	Origin	Schedulable $M=2H$?
REGCA1	Grid- <i>following</i> (current src.)	WECC/PSS/E REGC_A	no
REGCP1	Grid-forming (power/droop)	ANDES	no (M)
REGCV1	Grid-forming VSG (swing eq.)	ANDES (CURENT)	yes (M, D)
REGCV2	Grid-forming VSG (lag inner loop)	ANDES	yes (M, D)
REGF1	Grid-forming droop	EPRI GFM	no
REGF2	Grid-forming VSM	EPRI GFM	partial (m_f)
REGF3	Grid-forming dVOC	EPRI GFM	no

virtual rotor speed deviation $\Delta\omega$ and angle δ ,

$$M \frac{d\Delta\omega}{dt} = P'_{\text{ref}} - P_e - D \Delta\omega, \quad P'_{\text{ref}} = P_{\text{ref}} - k_\omega \Delta\omega, \quad (3)$$

$$\frac{d\delta}{dt} = 2\pi f_n \Delta\omega, \quad \omega = 1 + \Delta\omega, \quad (4)$$

with an inner double-loop PI voltage/current controller ($V_d, V_q \rightarrow I_d, I_q$ through r_a, x_s) closing the electrical dynamics. The ANDES parameters carry the exact physical meaning we need: M = “emulated startup time constant ($M=2H$)” (s) and D = “emulated damping coefficient” (p.u.), with k_ω the active-power (speed) droop.

Equation (3) is (1) with $M=2H$: at the instant of an imbalance, $d\Delta\omega/dt|_{0+} = \Delta P/M$, reproducing (2). Thus *the scheduler’s decision variable M is literally the inertia constant in the simulator’s swing equation*, and the surrogate learns the physically-grounded map $M \uparrow \Rightarrow |\dot{f}| \downarrow$. We verified this causally in the pipeline: with a fixed disturbance, raising the IBR inertia from $M=1$ to $M=7$ reduced $|\text{RoCoF}|$ by $1.45\times$, consistent with (2).

4 Why not the alternatives

REGCA1 (grid-following).

A current source synchronised by a PLL; it has no swing state and no inertia parameter (its parameters are current-limit and low-voltage logic). It *cannot* provide schedulable inertia and is therefore unusable for VIS. It is, however, the right *grid-following baseline* if one later wants a GFM-vs-GFL comparison.

REGF1 (droop), REGF3 (dVOC).

Grid-forming but inertia-free: REGF1 exposes droop gains (w_{drp}, Q_{drp}), REGF3 an oscillator. Neither has an $M=2H$ parameter, so “scheduling virtual inertia” is undefined for them. They address a different control question (droop/oscillator sharing), not inertia allocation.

REGF2 (EPRI VSM).

The strongest alternative: a virtual synchronous machine with an inertia-like coefficient m_f , damping d_d , a PLL, and explicit current/power limits ($P_{\text{max}}, Q_{\text{max}}, \text{PQFLAG}$). It is more standards-aligned (EPRI/UNIFI GFM) and more realistic. We nonetheless retain REGCV1 as the primary model because: (a) its $M=2H$ maps *directly and cleanly* to the VIS decision variable and to (1), whereas m_f requires an extra mapping; (b) its smaller state set and absence of hard current-limit switching keep the exact ReLU–MILP surrogate embedding cleaner; and (c) it preserves comparability with the foundational VIS work of She *et al.* (2024), from the same CURENT/ANDES group, which uses this model. REGF2 is the appropriate model for a future *realism robustness study*—in particular to add explicit converter current-limit dynamics as a predicted metric.

REGCV2 (VSG, lag inner loop).

Physically equivalent to REGCV1 with the inner current PI replaced by lag transfer functions; it carries the same M, D . It is an acceptable substitute; REGCV1 is the canonical, slightly simpler choice and the one used upstream.

5 Recommendation

1. **Primary model:** REGCV1. Decision-variable match ($M=2H, D$), swing-equation transparency (3) $\equiv$ (1), clean embeddability, and comparability with the reference VIS literature. This is the model used throughout the current pipeline and case files (`ieee39_full_ibrs.xlsx`), and the physics has been validated on the 50 Hz base.
2. **Robustness study (future):** REGF2. Re-run a subset with the EPRI VSM to confirm the conclusions are not model-specific and to introduce converter current-limit dynamics.
3. **Baseline (optional):** REGCA1. For an explicit grid-forming-vs-grid-following inertia comparison.

Bottom line: no model change is needed before the cluster campaign; REGCV1 is the correct and defensible choice for virtual inertia scheduling.

References

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